ASU Macro Theory Qualifier Playbook

Applied methodology from the 2016–2024 past qualifiers and 2026 problem-set emphasis

Built for use while working old qualifier problems - not a lecture-note substitute

How to use this document. Keep this next to the old exams. When you get stuck, do not search by topic name first. Search by *task*: "derive law of motion," "write stochastic Euler," "define RCE," "prove cutoff," or "answer Economist X." This playbook gives the exam-facing method, the equations to write, the common trap, and the old problems that train the move.

Contents

1 Exam DNA: the second-pass taxonomy

The supplied files contain the ASU Macro Theory qualifiers from 2016 through 2024. The early exams have optional fourth questions; the post-2018 format is three required problems with explicit instructions to allocate time to all three. The exam has become more applied, but the underlying moves are stable.

Rank	Module	Hits	Past problems	Core method
1	OLG and lifecycle saving	9	2016 Q2; 2017 Q3; 2018 Q2; 2019 Q2; 2020 Q3; 2021 Q1; 2022 Q2; 2023 Q1; 2024 Q2	Log lifetime budget; saving rule; aggregate saving; divide by next-period labor; steady-state comparison.
2	Dynamic programming and growth	9	2016 Q1; 2017 Q2; 2018 Q3; 2019 Q3; 2020 Q1; 2021 Q3; 2022 Q1; 2023 Q3; 2024 Q1	State; Bellman; FOC; envelope; Euler; log guess-and-verify; cutoff.
3	Asset pricing	6	2016 Q3; 2017 Q1; 2018 Q1; 2019 Q1; 2021 Q2; 2023 Q2	Recursive portfolio problem; FOC; envelope; stochastic discount factor; market clearing; log closed form.
4	Policy, welfare, wedges, Ramsey	7+	2016 Q1/Q4; 2017 Q2/Q4; 2020 Q1/Q3; 2021 Q2; 2024 Q3	Planner vs CE; implementability; externality wedge; tax changes price vs allocation.
5	RCE and distributions	1 direct, high current emphasis	2024 Q3; 2026 PS4	Individual state; aggregate state; value functions; prices; explicit distribution law; market clearing.
6	Search and reservation wages	2 direct, high current emphasis	2020 Q2; 2022 Q3; 2026 PS3	Accept value vs reject value; prove monotonicity; define cutoff; compare reservation wages.

Signature phrases to recognize immediately.

- "Formulate the problem recursively" means: state variables, Bellman, constraints, transition, policy rules.
- "Derive FOCs and Envelope" means: write the Bellman first, then FOC, then envelope, then Euler.
- "Evaluated at equilibrium allocations and prices" means: derive the household Euler before imposing market clearing.
- "Define a recursive competitive equilibrium" means: objects plus conditions plus an explicit law of motion for the distribution.
- "Economist X claims" means: answer with one formal equation and one intuition sentence. No essays.

1.1 How the exam has evolved

2016–2017: broad canonical theory. Arrow-Debreu equilibrium, Ramsey taxation, endogenous growth externalities, standard Lucas trees, OLG with skill or public infrastructure.

2018–2021: same canonical methods, but with timing and interpretation traps: time-varying asset supply, three-period OLG with survival, two-country asset markets, public leisure, search with assets, participation thresholds, and asset taxes.

2022–2024: applied dynamic choices. Safe vs risky project, home production, temporary UI, bond liquidity, alternating management, disaster risk, balanced growth with home goods, occupational choice with switching costs.

2026 problem sets emphasize four living skills: dynamic programming and verification, Lucas-tree pricing, search cutoffs, and RCE definitions with explicit distribution laws.

2 The first-round exam algorithm

The 4-line minimum answer. On any problem, before doing algebra, try to put these four lines on paper:

1. Individual or planner problem.
2. FOC plus envelope or accept/reject comparison.
3. Market clearing, government budget, or distribution law.
4. One sentence of intuition for the sign or comparison.

If those four lines are present, you are in the points. If any one is missing, fix that before polishing algebra.

2.1 Five-minute classification pass

Read all three questions and mark them by method:

If you see	**Start by writing**
Two-period agents, young/old, pension, home hours	Lifetime budget and log saving rule.
Tree, dividend, stock, bond, taxes	Recursive portfolio problem and Euler price equation.
Capital law, disaster, project, resource, skill	Bellman, FOC, envelope, Euler.
Continuum with idiosyncratic shocks	RCE objects and distribution law.
Wage offers, UI, unemployed/employed states	Accept value, reject value, reservation wage equality.
Government taxes/spending, externality	Planner/CE wedge or implementability constraint.

2.2 Time allocation

For a three-problem exam, use approximately 50–55 minutes per problem plus a final 10–15 minute rescue pass. Never spend 80 minutes chasing a closed form if another problem still lacks a Bellman equation or market-clearing condition.

2.3 Notation discipline

Use distinct symbols:

- Hours worked: h or n. Leisure: ℓ. Do not use l for both.
- Gross return: $R_{t+1} = 1 + r_{t+1}$. If production has depreciation, $R_{t+1} = 1 - \delta + F_K(K_{t+1}, L_{t+1})$.
- Capital-labor ratio: define the denominator every time, for example $k_t = K_t/L_t$ or $k_t = K_t/(X_t N_t n_t)$.

3 Universal templates

3.1 Template A: Bellman, FOC, envelope, Euler

Use this whenever the problem says "set up recursively," "derive Euler," "interpret optimal savings," or "guess and verify."

Resource constraint:

$$c + k' = f(k, z) + (1 - \delta)k.$$

Bellman equation:

$$V(k, z) = \max_{0 \le k' \le f(k,z)+(1-\delta)k} \left\{ u\big(f(k, z) + (1 - \delta)k - k'\big) + \beta \mathbb{E}[V(k', z') \mid z] \right\}.$$

FOC:

$$u'(c) = \beta \mathbb{E}[V_k(k', z') \mid z].$$

Envelope:

$$V_k(k, z) = u'(c)\big(f_k(k, z) + 1 - \delta\big).$$

Euler:

$$\boxed{u'(c_t) = \beta \mathbb{E}_t \left[u'(c_{t+1})\big(f_k(k_{t+1}, z_{t+1}) + 1 - \delta\big) \right].}$$

Trap. Do not jump directly to the Euler equation. On ASU exams, "derive" means the grader expects the FOC and envelope displayed separately.

3.2 Template B: stochastic asset-pricing Euler

Household with one tree:

$$V(a, s) = \max_{c,a'} \{u(c) + \beta \mathbb{E}[V(a', s') \mid s]\}$$

subject to

$$c + p(s)a' = [p(s) + d(s)]a.$$

FOC:

$$p(s)u'(c) = \beta \mathbb{E}[V_a(a', s') \mid s].$$

Envelope:

$$V_a(a, s) = u'(c)[p(s) + d(s)].$$

Euler:

$$\boxed{p(s)u'(c) = \beta \mathbb{E} \left[u'(c')\big(p(s') + d(s')\big) \mid s \right].}$$

Equilibrium with fixed supply one:

$$a = a' = 1, \qquad c = d(s).$$

Pricing formula:

$$\boxed{p(s) = \mathbb{E} \left[M(s, s')\big(p(s') + d(s')\big) \mid s \right], \qquad M(s, s') = \beta \frac{u'(c(s'))}{u'(c(s))}.}$$

3.3 Template C: two-period log OLG saving

Young resources e_1, old resources e_2, gross return R:

$$c_1 + s = e_1, \qquad c_2 = Rs + e_2.$$

Lifetime budget:

$$c_1 + \frac{c_2}{R} = e_1 + \frac{e_2}{R}.$$

Log FOC:

$$\frac{1}{c_1} = \beta R \frac{1}{c_2}.$$

Closed-form saving:

$$\boxed{s = \frac{\beta}{1+\beta} e_1 - \frac{1}{1+\beta}\frac{e_2}{R}.}$$

Use this before thinking. Then aggregate.

3.4 Template D: reservation wage or cutoff

Define accept value $A(w)$ and reject value R. If $A(w)$ is strictly increasing in w and R does not depend on w, then there is a cutoff w^* satisfying

$$\boxed{A(w^*) = R.}$$

The policy is

$$\text{accept iff } w \geq w^*.$$

For a permanent wage job with risk-neutral utility:

$$A(w) = \frac{w}{1-\beta}, \qquad w^* = (1-\beta)U.$$

3.5 Template E: RCE with explicit distribution law

Individual state x, aggregate state $S = (Z, \mu)$, policy $g(x, S, \varepsilon')$, shock transition Q. The law of motion for the distribution is

$$\boxed{\mu'(B) = \int_X \Pr\{g(x, S, \varepsilon') \in B \mid x, S\}\, d\mu(x).}$$

If the next state is deterministic conditional on next shock:

$$\boxed{\mu'(B) = \int_X \int_{\varepsilon'} \mathbf{1}\{g(x, S, \varepsilon') \in B\} Q(d\varepsilon' \mid x, S) d\mu(x).}$$

A recursive competitive equilibrium consists of:

1. value functions;
2. policy functions;
3. price functions;
4. firm policies;
5. government rules, if any;
6. aggregate law of motion;
7. distribution law;
8. market clearing.

Trap. "Prices depend only on aggregate K" does not imply K is the only aggregate state. Future aggregate capital is $K' = \int g(k, \theta, S) d\mu$, so the distribution may still be needed to forecast the future.

4 Playbook 1: OLG and lifecycle saving

4.1 The OLG engine

Every OLG problem uses the same engine:

1. Write the household lifetime budget.
2. Solve individual saving, usually with log utility.
3. Aggregate savings across types/cohorts.
4. Impose $K_{t+1} = S_t$.
5. Divide by the correct next-period labor input.
6. Solve steady state or compare coefficients.

The formula to memorize.

$$s_t^i = \frac{\beta}{1+\beta} e_{1t}^i - \frac{1}{1+\beta} \frac{e_{2,t+1}^i}{R_{t+1}}.$$

Young income raises saving. Old-age income, pensions, and high future labor income lower saving.

4.2 Capital and wages

For Cobb-Douglas $Y = K^\alpha L^{1-\alpha}$, define $k = K/L$. Then

$$w_t = (1-\alpha)k_t^\alpha, \qquad R_{t+1} = 1 - \delta + \alpha k_{t+1}^{\alpha-1}.$$

If capital fully depreciates, $R_{t+1} = \alpha k_{t+1}^{\alpha-1}$.

With no old-age resources and population growth n:

$$s_t = \frac{\beta}{1+\beta} w_t,$$

so

$$\boxed{k_{t+1} = \frac{1}{1+n} \frac{\beta}{1+\beta} (1-\alpha) k_t^\alpha.}$$

4.3 Type heterogeneity: the aggregate labor-income trick

If type i has young labor income $w_t e_i$ and no old income, then

$$s_t^i = \frac{\beta}{1+\beta} w_t e_i.$$

Aggregate saving is

$$K_{t+1} = \frac{\beta}{1+\beta} w_t L_t = \frac{\beta}{1+\beta} (1-\alpha) Y_t.$$

This is why many type-composition changes scale both saving and effective labor. They may raise total GDP, but not necessarily the steady-state capital-labor ratio or the return on capital.

4.4 Skilled/unskilled premium method

With

$$F(K, S, U) = K^\alpha L^{1-\alpha}, \qquad L = S^\phi U^{1-\phi},$$

we have

$$F_S = (1-\alpha)\phi \frac{Y}{S}, \qquad F_U = (1-\alpha)(1-\phi)\frac{Y}{U}.$$

Labor earnings per worker are $F_S z_s$ and $F_U z_u$. Therefore

$$\frac{F_S z_s}{F_U z_u} = \frac{\phi}{1-\phi}\frac{U}{S}\frac{z_s}{z_u}.$$

If $S = \pi z_s N$ and $U = (1-\pi) z_u N$, then

$$\boxed{\frac{F_S z_s}{F_U z_u} = \frac{\phi(1-\pi)}{(1-\phi)\pi}.}$$

A higher z_s does not mechanically raise the skill premium. It raises the supply of skilled efficiency units, lowering their marginal product. In the Cobb-Douglas aggregator, the effects cancel.

4.5 Social Security method

Young budget:

$$c_t^y + s_t = (1-\tau)w_t.$$

Old budget:

$$c_{t+1}^o = R_{t+1}s_t + p_{t+1}.$$

Balanced PAYG budget:

$$N_t^y \tau w_t = N_t^o p_t.$$

With $N_t^y = (1+n)N_t^o$,

$$p_t = (1+n)\tau w_t.$$

Saving is

$$\boxed{s_t = \frac{\beta}{1+\beta}(1-\tau)w_t - \frac{1}{1+\beta}\frac{p_{t+1}}{R_{t+1}}.}$$

Social Security lowers capital because it lowers young disposable wage income and adds pension wealth that substitutes for private saving.

4.6 Old-age endowment or loan economy

If type i receives z^i when young and $z^i(1-\delta^i)$ when old, then

$$s^i = \frac{z^i}{1+\beta}\left[\beta - \frac{1-\delta^i}{R}\right].$$

Loan-market clearing $\sum_i \omega_i s^i = 0$ gives

$$\boxed{R = \frac{\sum_i \omega_i z^i(1-\delta^i)}{\beta \sum_i \omega_i z^i}.}$$

If all δ^i's are equal, each type has the same proportional endowment profile and there is no trade.

4.7 Home production and hours

Let young time be split between market hours n and home hours h: $n + h = 1$. If market income is wn and there is no old-age income, then

$$s = \frac{\beta}{1+\beta} wn.$$

The key is that saving is still a fixed fraction of market labor income.

If utility contains $\log c_1 + \gamma \log(h^\lambda) + \beta \log c_2$, the optimal home hours satisfy

$$\frac{\gamma\lambda}{h} = \frac{1+\beta}{1-h},$$

so

$$\boxed{h^* = \frac{\gamma\lambda}{1+\beta+\gamma\lambda}, \qquad n^* = \frac{1+\beta}{1+\beta+\gamma\lambda}.}$$

If home output is $c_2 = Ah$ and period utility is

$$v(c_1, c_2) = \log c_1 + \frac{c_2^{1-\sigma}}{1-\sigma},$$

then the market/home FOC is

$$\boxed{\frac{1+\beta}{n} = A^{1-\sigma}(1-n)^{-\sigma}.}$$

Higher A lowers market hours if $\sigma < 1$, has no effect if $\sigma = 1$, and raises market hours if $\sigma > 1$.

4.8 Participation threshold

If a two-person household gets income w with one worker and $w(1+h)$ with two workers, but pays cost q, then with household utility $2[\log c_1 + \beta \log c_2] - q$, the gain from two workers is

$$2(1+\beta)\log(1+h) - q.$$

Thus

$$\boxed{q^* = 2(1+\beta)\log(1+h).}$$

Both work iff $q \le q^*$. Since prices cancel, the threshold is time invariant.

4.9 Balanced-growth labor denominator

If market labor efficiency X_t grows at gross rate $1+g$, population grows at gross rate $1+n$, and market hours are n_t, define

$$k_t = \frac{K_t}{X_t N_t n_t}.$$

Then a typical log-saving law becomes

$$\boxed{k_{t+1} = \frac{\beta}{1+\beta} \frac{1-\alpha}{(1+n)(1+g)} \frac{n_t}{n_{t+1}} k_t^\alpha.}$$

The ratio n_t/n_{t+1} is the whole transition after a home-productivity shock. Do not say "balanced growth path" makes hours constant after a permanent shock. BGP describes the steady path, not the impact response.

4.10 OLG practice map

Problem	**Method to drill**
2016 Q2	Skilled/unskilled labor, skill premium, aggregate labor-income trick.
2017 Q3	Public infrastructure, labor tax, government budget, steady-state private capital.
2018 Q2	Three-period OLG, cohort asset positions, survival/death probability.
2019 Q2	Old-age labor endowments, borrowing by high future earners, dynamic efficiency.
2020 Q3	Social Security, pension wealth, capital-law coefficient.
2021 Q1	Participation threshold and log indirect utility.
2022 Q2	Home production, type composition, GDP per worker versus return.
2023 Q1	Endowment-loan OLG, interest-rate clearing, no-trade case.
2024 Q2	Home goods, labor-efficiency growth, market-hours transition.

5 Playbook 2: Asset pricing

5.1 The asset-pricing engine

Write the problem before imposing equilibrium:

$$V(a,b,s) = \max_{c,a',b'} \{u(c) + \beta \mathbb{E}[V(a',b',s') \mid s]\}$$

subject to

$$c + p(s)a' + q(s)b' = [p(s) + d(s)]a + b + T(s).$$

Stock Euler:

$$p(s)u'(c) = \beta \mathbb{E}[u'(c')(p(s') + d(s')) \mid s].$$

Bond Euler:

$$q(s)u'(c) = \beta \mathbb{E}[u'(c') \mid s].$$

Then impose market clearing.

5.2 Log one-tree shortcut

If $u(c) = \log c$ and the tree is the only consumption good, then $c = d(s)$. Guess $p(s) = \eta d(s)$. The Euler gives

$$\eta = \beta(\eta + 1),$$

so

$$\boxed{p(s) = \frac{\beta}{1-\beta} d(s).}$$

This works even with Markov dividends because consumption equals dividends.

5.3 Two trees with constant aggregate consumption

If $d_1(s) + d_2(s) = 1$, then marginal utility is constant. Prices solve

$$p_j(s) = \beta \mathbb{E}[p_j(s') + d_j(s') \mid s].$$

Adding across trees:

$$P(s) = p_1(s) + p_2(s) = \beta \mathbb{E}[P(s') + 1 \mid s].$$

The constant solution is

$$\boxed{p_1(s) + p_2(s) = \frac{\beta}{1-\beta}.}$$

Persistence raises the price sensitivity of the currently high-dividend tree because current high dividends forecast future high dividends.

5.4 Iid dividend plus constant tree

If $d_1 = x$, $d_2 = 1$, and x' is iid, then $c = 1 + x$. With log utility,

$$p_j(x) = \beta(1+x)\mathbb{E}\left[\frac{p_j(x') + d_j(x')}{1+x'}\right].$$

Guess $p_j(x) = A_j(1+x)$. Then

$$\boxed{A_j = \frac{\beta}{1-\beta}\mathbb{E}\left[\frac{d_j(x')}{1+x'}\right].}$$

Both prices rise with current x, even the constant-dividend tree, because current high consumption lowers current marginal utility.

5.5 Dividend or payoff taxes

If dividends are taxed at next-period rate τ', then

$$\boxed{p(s)u'(c) = \beta\mathbb{E}[u'(c')\{p(s') + (1-\tau')d(s')\} \mid s].}$$

If the exam taxes the entire next payoff $p(s') + d(s')$ at rate τ, then

$$p(s)u'(c) = \beta(1-\tau)\mathbb{E}[u'(c')(p(s') + d(s')) \mid s].$$

With log utility and one tree:

$$\boxed{p(s) = \frac{\beta(1-\tau)}{1-\beta(1-\tau)}d(s).}$$

Lump-sum rebates can leave the real allocation unchanged while asset prices change. Do not infer inefficiency from a price wedge alone.

5.6 Growing deterministic dividend

If $d_{t+1} = \gamma d_t$, $\gamma > 1$, and utility is CRRA with coefficient σ, guess $p_t = \eta d_t$. The Euler gives

$$\eta = \beta\gamma^{1-\sigma}(\eta + 1),$$

so

$$\boxed{p_t = \frac{\beta\gamma^{1-\sigma}}{1-\beta\gamma^{1-\sigma}}d_t.}$$

The condition for a finite price-dividend ratio is $\beta\gamma^{1-\sigma} < 1$.
Return:

$$\boxed{R_{t+1} = \frac{\gamma^{\sigma}}{\beta}.}$$

A higher growth rate raises the asset price only if $\sigma < 1$. If $\sigma > 1$, growth makes future consumption high and future marginal utility low, which can lower the price-dividend ratio.

5.7 War or disaster risk

After war, dividends are sz. The after-war price is

$$\boxed{p^A = \frac{\beta}{1-\beta}sz.}$$

Before war, if war arrives with probability π, CRRA price $p^B = \eta z$ satisfies

$$\boxed{\eta = \frac{\beta(1-\pi) + \frac{\beta\pi s^{1-\sigma}}{1-\beta}}{1-\beta(1-\pi)}.}$$

Compare to $\beta/(1-\beta)$. The sign depends on $s^{1-\sigma} - 1$:

$$\eta \begin{cases} < \beta/(1-\beta), & \sigma < 1, \\ = \beta/(1-\beta), & \sigma = 1, \\ > \beta/(1-\beta), & \sigma > 1. \end{cases}$$

With log utility, the fall in dividends and the rise in marginal utility exactly offset.

5.8 Bond in utility

If utility is $u(c+\kappa b)$, then the inherited bond raises both resources and the utility argument. The envelope for the bond is

$$V_b = (1+\kappa)u'(c+\kappa b).$$

Bond Euler in equilibrium $b = 0$:

$$\boxed{q(s)u'(d(s)) = \beta(1+\kappa)\mathbb{E}[u'(d(s')) \mid s].}$$

With log utility:

$$\boxed{q(s) = \beta(1+\kappa)d(s)\mathbb{E}\left[\frac{1}{d(s')} \mid s\right].}$$

The stock price is unchanged relative to the standard log tree. Larger κ raises the bond price and lowers the risk-free rate, so the equity premium rises.

5.9 Time-varying net supply

Budget with beginning-of-period holdings y_{t-1}:

$$c_t + p_t y_t = (p_t + z)y_{t-1}.$$

Market clearing:

$$y_t = \bar{y}_t.$$

Euler:

$$p_t U'(c_t) = \beta U'(c_{t+1})(p_{t+1} + z).$$

With linear utility, U' is constant, so

$$p_t = \beta(p_{t+1} + z).$$

In the alternating even/odd environment, $p_E = p_O = \beta z/(1-\beta)$. With strict concavity, the state in which agents must absorb more shares has lower consumption and higher marginal utility, so the price is lower.

5.10 Two-country stock and common bond

Country i can trade only domestic stock but all countries can trade the bond. Country i's Euler equations are

$$p_i(s)U'(c_i) = \beta\mathbb{E}[U'(c_i')(p_i(s') + z_i') \mid s],$$

$$q(s)U'(c_i) = \beta\mathbb{E}[U'(c_i') \mid s].$$

Bond market clearing requires $b_1 + b_2 = 0$. Stock market clearing requires each country holds its domestic tree. If the proposed equilibrium has constant consumption, the bond price is $q = \beta$, and stock prices become discounted payoff sums under constant marginal utility.

5.11 Asset-pricing practice map

Problem	Method to drill
2016 Q3	Two trees, recursive pricing, log iid closed forms.
2017 Q1	Negatively correlated trees, constant aggregate consumption, persistence.
2018 Q1	Time-varying net supply and beginning-of-period holdings.
2019 Q1	Two countries, restricted stock trade, common bond.
2021 Q2	Asset tax, rebates, allocation versus price.
2023 Q2	Stock and bond with bond-in-utility liquidity value.
2026 PS2	Taxes, growth, war risk, CRRA/log comparisons.

6 Playbook 3: Dynamic programming and growth

6.1 Log/Cobb-Douglas full-depreciation shortcut

If

$$c + k' = zk^{\alpha}, \qquad u(c) = \log c,$$

then guess

$$k' = szk^{\alpha}, \qquad c = (1-s)zk^{\alpha}.$$

Euler:

$$\frac{1}{c_t} = \beta \mathbb{E}_t \left[\frac{\alpha z_{t+1} k_{t+1}^{\alpha-1}}{c_{t+1}} \right].$$

Under the guess,

$$\frac{\alpha z_{t+1} k_{t+1}^{\alpha-1}}{c_{t+1}} = \frac{\alpha}{(1-s)k_{t+1}}.$$

Thus $s = \alpha\beta$, and

$$\boxed{k' = \alpha\beta z k^{\alpha}, \qquad c = (1-\alpha\beta) z k^{\alpha}.}$$

6.2 AK log shortcut

If

$$c + k' = Ak, \qquad u(c) = \log c,$$

then guess $V(k) = A_0 + B \log k$. Coefficient matching gives $B = 1/(1-\beta)$. The FOC gives

$$\boxed{k' = \beta Ak, \qquad c = (1-\beta)Ak.}$$

6.3 Disaster AK with CRRA

Disaster occurs before production. Let post-disaster capital be x, where $x = k$ with probability $1-q$ and $x = \rho k$ with probability q. Resource:

$$c + k' = Ax.$$

Guess

$$k' = mAx.$$

Euler implies

$$1 = \beta A^{1-\sigma} m^{-\sigma} \left[(1-q) + q\rho^{1-\sigma} \right].$$

Hence

$$\boxed{m = \left\{ \beta A^{1-\sigma} \left[(1-q) + q\rho^{1-\sigma} \right] \right\}^{1/\sigma}.}$$

If $\sigma > 1$, a higher disaster probability raises saving. If $\sigma < 1$, it lowers saving. If $\sigma = 1$, disaster risk does not affect the saving share.

6.4 Natural resource extraction

Stock law:

$$x_{t+1} = (x_t - l_t)(1+g).$$

Consumption:

$$c_t = f(l_t).$$

Euler:

$$\boxed{u'(f(l_t))f'(l_t) = \beta(1+g)u'(f(l_{t+1}))f'(l_{t+1}).}$$

If $u(c) = \log c$ and $f(l) = l^\gamma$, then $u(f(l)) = \gamma \log l$. Guess $V(x) = A + B\log x$. Coefficient matching gives $B = \gamma/(1-\beta)$, and the FOC gives

$$\boxed{l_t = (1-\beta)x_t.}$$

Then

$$x_{t+1} = \beta(1+g)x_t,$$

and consumption is constant iff

$$\boxed{\beta(1+g) = 1.}$$

6.5 Skill accumulation

Budget:

$$c + i = ws.$$

Skill law:

$$s' = (1-\delta)s + As^{1-\gamma}i^\gamma.$$

Bellman:

$$V(s) = \max_{0\le i\le ws} \left\{u(ws-i) + \beta V((1-\delta)s + As^{1-\gamma}i^\gamma)\right\}.$$

FOC:

$$u'(c) = \beta V_s(s')A\gamma s^{1-\gamma}i^{\gamma-1}.$$

Envelope:

$$V_s(s) = u'(c)w + \beta V_s(s')\left[(1-\delta) + A(1-\gamma)s^{-\gamma}i^\gamma\right].$$

With log utility, guess $i = xs$ and $V(s) = A_0 + B\log s$. Then $B = 1/(1-\beta)$ and x solves

$$\boxed{\frac{1}{w-x} = \frac{\beta}{1-\beta}\frac{A\gamma x^{\gamma-1}}{1-\delta+Ax^\gamma}.}$$

6.6 Safe/risky project cutoff

For fixed project i, alive-state value:

$$V_i(k) = \max_{0 \le k' \le k} \left\{ \log(k - k') + \beta[\gamma_i V_i(A_i k') + (1 - \gamma_i)B] \right\}.$$

Guess $V_i(k) = a_i + m_i \log k$. Coefficient matching:

$$m_i = 1 + \beta\gamma_i m_i,$$

so

$$m_i = \frac{1}{1 - \beta\gamma_i}.$$

FOC gives

$$\boxed{k_i' = \beta\gamma_i k, \qquad c_i = (1 - \beta\gamma_i)k.}$$

If $\gamma_S > \gamma_R$, then safe project has higher continuation elasticity m_i. Therefore $V_S(k) - V_R(k)$ is increasing in $\log k$, giving a cutoff $\bar{k}$. Higher wealth favors the safer project.

6.7 Alternating management

Independent AK-log management gives

$$K_{t+1} = \beta A K_t.$$

Alternating management Bellman:

$$V_i(k) = \max_{c, c', s, k'} \{\log c + \beta \log c' + \beta^2 V_i(g_{-i}(k'))\}$$

subject to

$$c + s + k' = Ak, \qquad c' = As.$$

In a symmetric linear equilibrium, the share put into joint capital is β^2, so

$$\boxed{K_{t+1} = \beta^2 A K_t.}$$

Alternating management grows more slowly because the current manager internalizes only her own consumption and delayed continuation access.

6.8 Public leisure externality

Planner internalizes that $\ell = \bar{\ell}$. Competitive households take $\bar{\ell}$ as given. Planner labor FOC:

$$u_c(c) F_h(K, h) = \phi_1(\ell, \ell) + \phi_2(\ell, \ell).$$

Competitive FOC:

$$u_c(c) w = \phi_1(\ell, \bar{\ell}).$$

With $\phi_2 > 0$, CE works too much.
In the log/Cobb-Douglas/full-depreciation case with utility

$$\log c + b_1 \log \ell + b_2 \log \bar{\ell},$$

steady-state planner hours are

$$\boxed{h^{SP} = \frac{1 - \gamma}{(1 - \gamma) + (b_1 + b_2)(1 - \beta\gamma)}.}$$

CE hours replace $b_1 + b_2$ by b_1, so $h^{CE} > h^{SP}$.

6.9 Dynamic programming practice map

Problem	Method to drill
2016 Q1	Pareto planner, aggregation, distribution of initial capital.
2017 Q2	Endogenous growth externality, private vs social return.
2018 Q3	Natural resource extraction and log closed form.
2019 Q3	Irreversible work decision with assets and cutoff in initial wealth.
2020 Q1	Public leisure planner, CE externality.
2021 Q3	Skill accumulation, FOC/envelope, linear investment guess.
2022 Q1	Safe/risky project, cutoff, survival risk.
2023 Q3	AK log, independent vs alternating management.
2024 Q1	Disaster AK, CRRA saving share.
2026 PS1	DP foundations, contraction, monotonicity, stochastic growth.

7 Playbook 4: RCE and distributions

7.1 The RCE checklist

For any RCE definition, write these objects explicitly:

1. Individual state x.
2. Aggregate state S, usually (Z, μ).
3. Value functions $V(x, S)$.
4. Policy functions $g(x, S)$.
5. Price functions $p(S)$, wages, interest rates, taxes.
6. Firm policies.
7. Government budget/rule, if present.
8. Distribution law $\mu' = H(S)$.
9. Market clearing.

7.2 Inelastic-labor capital economy with idiosyncratic taste shocks

State:

$$x = (k, \theta), \qquad S = \mu.$$

Aggregate capital:

$$K(\mu) = \int k \, d\mu(k, \theta).$$

Prices with $Y = K^\alpha N^{1-\alpha}$ and $N = 1$:

$$r(\mu) = \alpha K(\mu)^{\alpha-1}, \qquad w(\mu) = (1-\alpha)K(\mu)^\alpha.$$

Household Bellman:

$$V(k, \theta, \mu) = \max_{c,k'} \left\{ \theta u(c) + \beta \int V(k', \theta', \mu') Q(d\theta' \mid \theta) \right\}$$

subject to

$$c + k' = [1 - \delta + r(\mu)]k + w(\mu).$$

Distribution law:

$$\boxed{\mu'(B_k \times B_\theta) = \int \mathbf{1}\{g_k(k, \theta, \mu) \in B_k\} Q(\theta, B_\theta) d\mu(k, \theta).}$$

Even though r depends only on K, the aggregate state generally cannot be only K, because

$$K' = \int g_k(k, \theta, \mu) d\mu(k, \theta)$$

depends on the distribution.

7.3 Fixed skilled/unskilled population

State for type $i \in \{s, u\}$:

$$x_i = k_i.$$

Aggregate state:

$$S = (\mu_s, \mu_u)$$

or, since individuals within each group are identical in the simplest version, (k_s, k_u). Production:

$$F(K, N_s, N_u).$$

Prices:

$$r = F_K(K, N_s, N_u) - \delta, \quad w_s = F_{N_s}(K, N_s, N_u), \quad w_u = F_{N_u}(K, N_s, N_u).$$

Market clearing:

$$K = \theta k_s + (1-\theta)k_u,$$

$$N_s = \theta n_s, \qquad N_u = (1-\theta)n_u.$$

Prices depend on aggregate capital and type labor inputs, but if individual capital distributions within groups become nondegenerate, policy aggregation may require more than average capital.

7.4 Tradable versus nontradable home good

Good 2 home technology:

$$y_2 = \theta h.$$

Market wage in good 1: w. Price of good 2 in good 1: p_2.
If good 2 is tradable, time allocation solves a linear problem:

$$\max_{n,h\geq 0} \; wn + p_2\theta h \quad \text{s.t. } n + h = 1.$$

Thus:

$$h = \begin{cases} 1, & p_2\theta > w, \\ 0, & p_2\theta < w, \\ \text{any } h \in [0,1], & p_2\theta = w. \end{cases}$$

If good 2 is nontradable, then $c_2 = \theta h$. The labor choice is interior under Inada conditions and satisfies

$$\boxed{U_2(c_1, c_2)\theta = U_1(c_1, c_2)w.}$$

Economist X trap: nontradability does not by itself prove Pareto inefficiency. The planner facing the same nontradability constraint may still decentralize the allocation. Inefficiency requires a wedge relative to the relevant feasible set.

7.5 Sector-specific human capital

Individual state:

$$x = (h_A, h_B).$$

Aggregate state:

$$S = (z, \mu), \qquad z = z_A/z_B.$$

If the worker chooses sector i, income is $w_i(S)h_i$. Human capital evolves as

$$h_i' = (1+\varepsilon)h_i \quad \text{if working in } i,$$

$$h'_j = (1-\delta)h_j \quad \text{if not working in } j.$$

Bellman:

$$V(h_A, h_B, z, \mu) = \max_{i \in \{A,B\}} \left\{ u(w_i(S)h_i + \Pi(S)) + \beta \mathbb{E}[V(h'_A, h'_B, z', \mu') \mid z] \right\}.$$

Effective labor clearing:

$$H_i = \int_{\{x: g_i(x,S)=i\}} h_i \, d\mu(x).$$

Distribution law:

$$\mu'(B) = \int \mathbf{1}\{(h'_A(x), h'_B(x)) \in B\} d\mu(x),$$

combined with the Markov transition for z.

7.6 Occupational choice with entrepreneurs and switching costs

Individual state:

$$x = (z, o_{-1}), \qquad o_{-1} \in \{W, E\}.$$

Aggregate state:

$$S = (G, \mu).$$

Entrepreneur profit:

$$\pi(z, w) = \max_n \{zf(n) - wn\}, \qquad zf'(n) = w.$$

Worker consumption:

$$c_W = w.$$

Entrepreneur consumption:

$$c_E = (1-\tau)\pi(z, w).$$

Bellman:

$$V(z, o_{-1}, S) = \max_{o \in \{W,E\}} \left\{ u(c_o(z,S)) - \gamma \mathbf{1}\{o \neq o_{-1}\} + \beta \mathbb{E}[V(z', o, S') \mid z, G] \right\}.$$

Government budget:

$$\boxed{G = \tau \int_{\{g_o(x,S)=E\}} \pi(z, w(S)) d\mu(x).}$$

Labor market clearing:

$$\boxed{\int_{\{g_o=E\}} n(z, w(S)) d\mu = \int_{\{g_o=W\}} 1 \, d\mu.}$$

Distribution law:

$$\boxed{\mu'(B_z \times B_o) = \int Q(z, B_z) \mathbf{1}\{g_o(z, o_{-1}, S) \in B_o\} d\mu(z, o_{-1}).}$$

With $\gamma = 0$, constant G, and iid ideas, a higher G requires higher τ. This lowers after-tax entrepreneurship value. At the old wage, too few entrepreneurs and too many workers want to enter the labor market. Equilibrium wage falls.

7.7 RCE practice map

Problem	**Method to drill**
2020 Q2	Search RCE with asset market and invariant distribution.
2024 Q3	Occupational choice, entrepreneurs, profit tax, switching costs.
2026 PS4 Q1	RCE with idiosyncratic taste shocks; explicit distribution law.
2026 PS4 Q2	Skilled/unskilled RCE and whether prices depend on averages.
2026 PS4 Q3	Tradable vs nontradable home good and efficiency claim.
2026 PS4 Q4	Sector-specific human capital and aggregate state.
2026 PS4 Q5	Heterogeneous firms, sector switching, adjustment cost.

8 Playbook 5: Search and reservation wages

8.1 Basic McCall search

Employed value:

$$W(w) = \frac{w}{1-\beta}.$$

Unemployed value:

$$U = b + \beta \int \max\{W(w), U\} dF(w).$$

Reservation wage:

$$\boxed{w^* = (1-\beta)U.}$$

Accept iff $w \geq w^*$.

8.2 Offer probability

If offers arrive with probability p:

$$U = b + \beta \left[(1-p)U + p \int \max\{W(w), U\} dF(w) \right].$$

A higher offer probability raises the option value of search, so U rises and w^* rises.

If the state affects only the probability of receiving an offer and the state is iid, then only the average arrival probability matters:

$$\bar{p} = \pi_h p_h + (1-\pi_h) p_\ell.$$

The reservation wage does not depend on the current state because once an offer arrives, the wage distribution is the same.

8.3 Stochastic benefits

High-benefit state:

$$U_H = \int \max \left\{ \frac{w}{1-\beta}, \bar{b} + \beta[pU_H + (1-p)U_L] \right\} dF(w).$$

Low state, absorbing:

$$U_L = \int \max \left\{ \frac{w}{1-\beta}, b + \beta U_L \right\} dF(w).$$

Reservation wages:

$$\boxed{\frac{w_H^*}{1-\beta} = \bar{b} + \beta[pU_H + (1-p)U_L], \qquad \frac{w_L^*}{1-\beta} = b + \beta U_L.}$$

Higher benefits raise the reject value and therefore raise reservation wages.

8.4 Temporary unemployment compensation with quitting

Use the states in the question:

- $v_e(w)$: employed last period at wage w, before quit decision.
- $v_1(w)$: unemployed, offer w, eligible for UI if rejecting.
- $v_+(w)$: unemployed, offer w, not eligible for UI.

Bellman equations:

$$v_e(w) = \max\left\{w + \beta v_e(w),\ b + \beta \int v_1(w')dF(w')\right\}.$$

$$v_1(w) = \max\left\{\beta v_e(w),\ b + \beta \int v_+(w')dF(w')\right\}.$$

$$v_+(w) = \max\left\{\beta v_e(w),\ \beta \int v_+(w')dF(w')\right\}.$$

Cutoffs exist because $v_e(w)$ is increasing in w.

Eligible unemployed workers reject more offers than ineligible workers:

$$\beta v_e(w_1^*) = b + \beta \mathbb{E} v_+,$$

$$\beta v_e(w_+^*) = \beta \mathbb{E} v_+.$$

Therefore:

$$\boxed{w_1^* > w_+^*.}$$

If both cutoffs lie on the employment-continuation branch, then

$$\boxed{w_1^* - w_+^* = \frac{1-\beta}{\beta} b.}$$

8.5 Search with assets

Reservation wage solves

$$V^e(a, w^*(a)) = V^u(a).$$

Implicit derivative:

$$w_a^*(a) = \frac{V_a^u(a) - V_a^e(a, w^*)}{V_w^e(a, w^*)}.$$

If employed workers consume more than unemployed workers at the same asset level, then $u'(c^e) < u'(c^u)$, so $V_a^e < V_a^u$. Hence

$$\boxed{w_a^*(a) > 0.}$$

Richer unemployed workers are more selective.

8.6 Temporary jobs

If jobs last one period and begin next period, accepting wage w today gives payoff next period and unemployment again afterward. The accept value is not $w/(1-\beta)$. It is typically

$$A_s(w) = \beta \left[w + \beta \mathbb{E} U_{s'}\right]$$

with the expectation determined by the state process. The reject value includes current UI and next-period unemployment value:

$$R_s = b + \beta \mathbb{E} U_{s'}.$$

Reservation wage in state s solves $A_s(w_s^*) = R_s$.

Timing trap: if employment starts next period, the current-period benefit is received only if rejecting. Do not use the permanent-job formula blindly.

8.7 Search practice map

Problem	**Method to drill**
2020 Q2	Search with assets, stationary RCE, reservation wage increasing in assets.
2022 Q3	Temporary UI, quit option, reservation-wage ranking.
2026 PS3 Q1	Offer probability and iid state.
2026 PS3 Q2	Stochastic unemployment benefits.
2026 PS3 Q3	Two offers and recall.
2026 PS3 Q4	Temporary jobs, iid vs Markov aggregate state.

9 Playbook 6: Policy, welfare, Ramsey, and efficiency

9.1 Pareto weights and aggregation

Planner with N households:

$$\max_{\{c_{nt}\}} \sum_n \lambda_n \sum_t \beta^t \log c_{nt}$$

subject to aggregate feasibility. FOC:

$$\frac{\lambda_n}{c_{nt}} = \mu_t.$$

Therefore:

$$\boxed{c_{nt} = \frac{\lambda_n}{\sum_j \lambda_j} C_t.}$$

The aggregate capital path depends on aggregate initial capital $K_0 = \sum_n k_{n0}$, not the distribution of K_0, under complete markets and homothetic preferences.

9.2 Ramsey implementability

Household labor FOC:

$$u_c(c_t, \ell_t)(1 - \tau_t^l)w_t = u_\ell(c_t, \ell_t).$$

Household capital Euler:

$$u_c(c_t, \ell_t) = \beta u_c(c_{t+1}, \ell_{t+1})[1 + (1 - \tau_{t+1}^k)(r_{t+1} - \delta)].$$

The Ramsey planner eliminates prices and taxes using household FOCs and budget constraints. Generic implementability:

$$\boxed{\sum_{t=0}^{\infty} \beta^t \left[u_c(c_t, \ell_t)c_t - u_\ell(c_t, \ell_t)\ell_t\right] = \text{initial wealth in marginal-utility units.}}$$

9.3 Zero capital tax after period 0

The standard result in the ASU Ramsey problem is

$$\boxed{\tau_t^k = 0 \quad \text{for } t \geq 1.}$$

Reason: a tax on future capital distorts intertemporal accumulation. A tax on initial capital, if allowed, is a wealth tax on a predetermined object. If the problem imposes $\tau_0^k = 0$, that does not justify distorting all future savings margins.

9.4 Labor tax smoothing without capital

Utility:

$$c_t - \frac{l_t^{1+\varphi}}{1+\varphi}.$$

Production $y_t = l_t$. Household FOC:

$$l_t^\varphi = 1 - \tau_t.$$

Tax revenue:

$$\tau_t l_t = (1 - l_t^{\varphi}) l_t.$$

Government intertemporal budget:

$$\boxed{\sum_t \beta^t g_t = \sum_t \beta^t (l_t - l_t^{1+\varphi}).}$$

Resource constraint $c_t = l_t - g_t$ gives welfare:

$$\boxed{\sum_t \beta^t \left[l_t - g_t - \frac{l_t^{1+\varphi}}{1+\varphi} \right].}$$

Tax smoothing beats period-by-period balanced budgets because distortion costs are convex in the tax wedge.

9.5 Endogenous growth externality

Firm technology:

$$y_i = \Phi k_i^{\theta} (A_t l_i)^{1-\theta}, \qquad A_t = K_t.$$

Firms take A_t as given. In symmetric equilibrium, private return is $\theta\Phi$, but social return is Φ. With log utility and full depreciation:

$$\frac{c_{t+1}^{CE}}{c_t^{CE}} = \beta\theta\Phi, \qquad \frac{c_{t+1}^{SP}}{c_t^{SP}} = \beta\Phi.$$

A capital subsidy satisfying

$$(1+s)\theta\Phi = \Phi$$

implements the social return:

$$\boxed{s = \frac{1-\theta}{\theta}.}$$

9.6 Efficiency diagnosis

Use this decision tree:

1. Does the household ignore a social effect? If yes, externality wedge.
2. Is there a missing market relative to the planner's feasible set? If yes, likely constrained inefficiency.
3. Is the asset in fixed supply with a representative household and lump-sum rebates? Price changes may not affect allocation.
4. Is a good nontradable? That changes the feasible set; it is not automatically inefficient.
5. Does a tax distort a margin actually chosen by households? Labor and future capital taxes usually do; fixed-supply asset taxes may not.

9.7 Policy practice map

Problem	**Method to drill**
2016 Q1	Pareto weights, complete markets, aggregation.
2016 Q4	Implementability and zero capital tax.
2017 Q2	Growth externality and capital subsidy.
2017 Q4	Labor-tax smoothing.
2020 Q1	Public leisure externality.
2021 Q2	Asset tax changes prices; allocation may remain efficient.
2024 Q3	Profit tax financing government spending and wage effects.

10 The "stuck on an old qual" rescue sheet

10.1 If the problem asks for a law of motion

Write:

$$K_{t+1} = \text{aggregate saving at } t.$$

Then write:

$$k_{t+1} = \frac{K_{t+1}}{\text{next-period effective labor}}.$$

If you cannot solve the entire coefficient, leave it as

$$k_{t+1} = C(\text{parameters})k_t^{\alpha}$$

and sign C formally.

10.2 If the problem asks for a closed form

Check for one of these gifts:

Gift	Guess
Two-period log OLG	$s = \frac{\beta}{1+\beta}e_1 - \frac{1}{1+\beta}e_2/R.$
Log Cobb-Douglas growth	$k' = \alpha\beta z k^{\alpha}.$
Log AK	$k' = \beta A k.$
Log Lucas tree, one tree	$p = \frac{\beta}{1-\beta}d.$
CRRA deterministic growth	$p_t = \eta d_t.$
Log value with homogeneous state	$V(x) = A + B\log x.$
Cutoff model	Solve equality of two values.

10.3 If the problem asks "Economist X claims"

Use this answer format:

Claim: Economist X is right/wrong/only partially right.
Formal reason: From equation $(*)$, the object moves because $\partial X/\partial p$ has sign $\cdot$.
Economic reason: The force is $\cdot$.
Trap: X confused $\cdot$ with $\cdot$.

Examples of common traps:

- More aggregate saving in levels does not imply lower return if effective labor rises too.
- Higher productivity of a type does not automatically raise that type's premium if relative supply changes.
- Tax rebates may undo resource effects but not payoff-price effects.
- BGP does not mean no transition after an unexpected permanent shock.
- Nontradability is a constraint, not automatically a Pareto inefficiency.

10.4 If the problem asks for an RCE

Write the object list first, even if you do not yet know the equations. Then fill equations:

$$\text{Household optimality}, \quad \text{firm optimality}, \quad \text{government budget}, \quad \text{market clearing}, \quad \mu' = H(\mu).$$

The distribution law is usually the easiest missing point to recover:

$$\mu'(B) = \int \Pr\{x' \in B \mid x, S\} d\mu(x).$$

10.5 If the problem asks for intuition

Use marginal-utility language, not stories. Examples:

- Asset price rises because current marginal utility is low relative to future marginal utility.
- Saving rises because future marginal utility in disaster states is high.
- Reservation wage rises because the option value of waiting rises.
- Competitive labor is too high because households ignore the public value of leisure.
- Wage falls after a profit-tax-financed government expansion because entrepreneurship becomes less attractive and labor supply rises relative to labor demand.

11 Problem crosswalk: all supplied exams

Problem	Label	Playbook move
2016 Q1	Multi-household growth and Pareto efficiency	Pareto weights; constant consumption shares; aggregate capital independent of distribution under complete markets.
2016 Q2	OLG skilled/unskilled	Log saving; effective labor; skill premium cancellation; steady-state return comparison.
2016 Q3	Two-tree asset pricing	Recursive portfolio problem; tree price functions; log iid closed form.
2016 Q4	Ramsey taxation	AD equilibrium; implementability; zero capital tax from period 1.
2017 Q1	Negatively correlated trees	Constant aggregate consumption; price-sum formula; persistence.
2017 Q2	Endogenous growth externality	Private vs social return; growth-rate comparison; corrective subsidy.
2017 Q3	OLG public infrastructure	Labor tax, government infrastructure capital, private capital law, claim critique.
2017 Q4	Optimal labor taxation	Household FOC; tax revenue; government PV budget; tax smoothing.
2018 Q1	Time-varying asset supply	Beginning holdings; market clearing timing; linear vs concave utility prices.
2018 Q2	Three-period OLG	Lifetime budget; cohort savings; capital-labor law with productivity growth and survival.
2018 Q3	Lumber/resource extraction	Sequential problem; Euler; log-power closed form; trade across growth rates.
2019 Q1	Two-country assets	Country-specific stocks, common bond, stationary RCE.
2019 Q2	OLG with age-varying labor	Old-age labor income lowers saving; borrowing by college type; efficiency claim.
2019 Q3	Irreversible work choice	Dynamic budget; borrowing constraint; consumption comparison; asset cutoff.
2020 Q1	Public leisure	Planner Bellman; public leisure wedge; steady-state hours.
2020 Q2	Search with assets	Stationary RCE; reservation wage increasing in assets.
2020 Q3	Social Security	PAYG budget; pension wealth; capital accumulation falls with tax.
2021 Q1	Two-person household OLG	Log indirect utility; participation threshold; output vs return.
2021 Q2	Lucas tree with tax	Taxed payoff Euler; log price; efficiency vs price.
2021 Q3	Skill accumulation	FOC/envelope; Euler; log linear investment rule.
2022 Q1	Project survival risk	Project-specific value; cutoff; log saving share $\beta\gamma_i$.
2022 Q2	Home production OLG	Hours FOC; type composition; market labor and capital-labor ratio.
2022 Q3	Temporary UI	Bellman states; reservation wage ranking; quit implications.
2023 Q1	Endowment-loan OLG	Interest-rate clearing; no trade with proportional profiles.
2023 Q2	Stock and bond liquidity	Bond-in-utility envelope; equity premium.
2023 Q3	AK management rights	Log AK saving; alternating management β^2 rule.
2024 Q1	Disaster AK	CRRA saving share; precautionary vs return effect.
2024 Q2	Home goods and BGP	Market/home FOC; effective labor denominator; permanent home productivity shock.
2024 Q3	Entrepreneur/worker RCE	Distribution law; profit tax; wage response to government spending.

12 Final one-page protocol

When you open a past qual:

1. Label each problem by method, not topic name.
2. Write the state variables immediately.
3. Write the primitive problem before any equilibrium substitution.
4. Derive FOC and envelope if dynamic; accept/reject equality if search; lifetime budget if OLG.
5. Impose market clearing only after the individual condition exists.
6. Use log gifts aggressively.
7. For any comparison, write the equation whose derivative/sign answers it.
8. End every part with one sentence of intuition.

The formulas to have in muscle memory.

$$s = \frac{\beta}{1+\beta}e_1 - \frac{1}{1+\beta}\frac{e_2}{R}, \qquad k' = \alpha\beta z k^{\alpha}, \qquad k' = \beta A k,$$

$$p(s)u'(c) = \beta\mathbb{E}[u'(c')(p(s') + d(s')) \mid s], \qquad p = \frac{\beta}{1-\beta}d,$$

$$\mu'(B) = \int \Pr\{x' \in B \mid x, S\} d\mu(x), \qquad A(w^*) = R.$$